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Global, regional and national disease burden of neonatal disorders from 2000 to 2021 and the comparison before and during COVID-19: a systematic analysis

Chenyuan Qin¹, Xiao Gao², Li Yin², Zhumei Fu², Xingang Sang³, Yansheng Ding² and Jue Liu^{1,4*}

Abstract

Background Infant and neonatal health is a critical indicator for assessing a country's well-being and development. This study aimed to systematically analyze the global burden, long-term trends, coronavirus disease 2019 (COVID-19) impactions and inequalities of neonatal disorders and individual causes between 2000 and 2021.

Methods We collected global, regional and national annual data of neonatal disorders and individual causes on incidence and mortality between 2000 and 2021. Estimated annual percentage changes (EAPCs) were used to assess long-term trends and the impact of COVID-19. Lorenz curve and concentration index (CCI) was performed to evaluate the inequalities of neonatal disorders across socio-demographic index (SDI) levels.

Results In 2021, there were 27,062,768 new cases and 1,831,535 deaths of neonatal disorders globally. Between 2000 and 2021, the incidence rate of neonatal disorders declined from 23,112.05 to 20,916.69 per 100,000 live births (EAPC = -0.63%), while the global mortality rate declined from 2048.34 to 1415.59 per 100,000 live births (EAPC = -1.76%). Since 2000, neonatal preterm birth has consistently had the highest incidence rate (16,658.81 in 2021, EAPC = -0.47%) and mortality rate (571.69 in 2021, EAPC = -1.73%). South Asia and sub-Saharan Africa were identified as high-burden regions. Globally, the incidence rate of neonatal disorders declined before the COVID-19 pandemic but began to rise during the pandemic, whereas the mortality rate continued to decline at a faster pace than the pre-pandemic period. Additionally, between 2000 and 2021, global burden of neonatal disorders was concentrated in lower SDI countries, particularly in mortality rates.

Conclusions Although global progress has been made in reducing the burden of neonatal disorders, Sub-Saharan Africa and South Asia remain high-burden regions requiring particular attention. To reduce global and regional disparities, particularly in the post-COVID-19 era, governments and international organizations are supposed to implement evidence-based strategies, such as increasing investments in maternal and neonatal care, expanding neonatal intensive care capacity, enhancing workforce training, improving health information systems, and addressing social determinants through integrated, multisectoral approaches.

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Keywords Neonatal disorders, Preterm birth, Neonatal encephalopathy, Neonatal sepsis, Hemolytic disease, COVID-19

Background

Infant and neonatal health — a critical indicator for assessing a country's well-being and development — is influenced by factors such as healthcare access, socioeconomic status, maternal education, environmental sanitation and nutrition [1–8]. The first year of life, especially during the neonatal period (the first 28 days), is marked by rapid growth and vulnerability, highly susceptible to diseases and injuries [9]. The primary causes of infant disease burden, particularly in low- and middle-income countries (LMICs), include premature birth and low birth weight, birth complications such as asphyxia and trauma, neonatal infections, neonatal jaundice, congenital anomalies, and malnutrition [10–12]. Infant mortality rate (IMR) is defined as the number of deaths among live-born infants before reaching one year of age per 1,000 live births, whereas neonatal mortality rate (NMR) refers specifically to deaths occurring within the first 28 days of life per 1,000 live births [13]. Globally, IMR has dropped from 61 per 1,000 live births in 1994 to 28 in 2019, marking a 54% reduction [14]. Children are at the highest risk of death in their first month of life, with an average NMR of 17 deaths per 1,000 live births in 2022, a 53% decrease from 37 per 1,000 in 1990 [15]. As of 2018, deaths within the first month of life accounted for 47% of all under-five mortality (U5MR) [14]. Approximately 2.3 million neonates died within the first 28 days of life in 2022, averaging about 6,300 neonatal deaths per day [15]. These deaths are often due to conditions linked to poor quality of care at birth or lack of skilled medical attention during and shortly after delivery [16]. These conditions are largely preventable through improved access to quality healthcare, skilled birth attendants, and timely medical interventions [12, 17]. It is concerning that these improvements in child survival have not been universally realized, especially in LMICs [18]. For instance, the average NMR and IMR were about 3.7 and 5.4 per 1,000 live births in North America, while these rates in Sub-Saharan Africa were much higher, with an NMR of 27.5 and an IMR of 51.7 per 1,000 live births in 2019 [19].

Since the 2000s, the international community has continued to promote public health interventions through multilateral cooperation to reduce infant mortality. Sustainable Development Goals (SDGs) aims to eliminate preventable child deaths by 2030, targeting NMR below 12 per 1,000 live births and U5MR below 25 per 1,000 live births [20]. This goal emphasizes equity, shifting from relative global targets (as outlined in the Millennium Development Goals) to absolute targets for each country

[21]. The World Health Organization (WHO) and United Nations International Children's Emergency Fund (UNICEF) have actively supported critical interventions, such as breastfeeding practices, postnatal visits and emergency obstetric care, which have shown initial success in high-risk regions like sub-Saharan Africa and South Asia [22–24]. However, challenges such as insufficient funding and weak health systems remain significant barriers. As of 2019, projections indicated that at least 65 countries (32%) — mostly in Sub-Saharan Africa and South Asia — are unlikely to meet any SDG 3.2 targets by 2030 [18, 25]. These regions, with the highest mortality risks and the lowest data availability, require urgent efforts to accelerate progress. Socioeconomic inequalities in maternal and child health persist, further exacerbated by the coronavirus disease 2019 (COVID-19) pandemic [26, 27]. Additionally, declining fertility rates and demographic shifts may lead to greater concentration of pregnancies and births in low-income regions, where healthcare infrastructure may be insufficient to meet neonatal demands [28, 29].

Thus, this study aimed to systematically analyze the global burden, long-term trends and inequalities of neonatal disorders and those individual causes between 2000 and 2021. Additionally, we also explored the impact of the COVID-19 pandemic on neonatal disorders. This study may provide data-driven insights to guide governments and international organizations in optimizing neonatal health interventions, thereby effectively reducing global and regional health inequalities.

Methods

Study design and data sources

The Global Burden of Disease Study 2021 (GBD 2021) database offers a comprehensive framework for assessing 371 diseases and injuries, 288 causes of death, and 88 risk factors at global, regional, and national levels [30]. Neonatal disorders were modelled as five individual causes: (1) neonatal preterm birth; (2) neonatal encephalopathy due to birth asphyxia and trauma; (3) neonatal sepsis and other neonatal infections; (4) hemolytic disease and other neonatal jaundice; (5) other neonatal disorders, which was included only for mortality estimates. All estimates were performed using standardized methodologies to ensure consistency across locations and time periods, and more details can be found in Supplementary Methods [30]. Since our study utilized secondary data aggregated at global levels, it was exempt from ethical approval.

Data extraction

We extracted data on neonatal disorders and the five individual causes of morbidity and mortality from 2000 to 2021, including incident cases, deaths, incidence rates (per 100,000 live births), and mortality rates (per 100,000 live births), along with their 95% uncertainty intervals (UIs). The socio-demographic index (SDI), which reflects the social and economic conditions influencing health outcomes, combines three key indicators: the total fertility rate for women under age 25, mean years of schooling for individuals aged 15 and older, and lag-distributed income per capita [30]. SDI scores range from 0 (lowest development) to 1 (highest development), and 204 countries and territories are categorized into five quintiles: Low SDI, low-middle SDI, middle SDI, high-middle SDI, and high SDI regions (Table S1). For administrative and analytical purposes, the world is further divided into 21 GBD regions—such as Andean Latin America, Australasia, Central Asia, and Central Europe—based on epidemiological similarities and geographic proximity [10].

Statistical analysis

We used relative percent changes and estimated annual percentage changes (EAPCs) to assess the trends in neonatal disorder-related health indicators from 2000 to 2021. Relative percent change was calculated as $(\text{Number_after} - \text{Number_before}) / \text{Number_before} \times 100\%$. EAPC, derived using $100 \times (\exp(\beta) - 1)$, was computed after confirming linear regression assumptions by $Y = \alpha + \beta X + \varepsilon$, where Y is the natural logarithm of the rate, X represents the calendar year, β indicates the trend, and ε accounts for error [31]. An upward trend in incidence or mortality rates during the study period was defined as both the EAPC and its 95% confidence interval (CI) > 0 , while a downward trend was defined as both being < 0 . We also compared the changes in burden between the pre-pandemic period (2000–2019) and the COVID-19 pandemic period (2019–2021). Socioeconomic inequalities were evaluated using the Lorenz curve and concentration index (CCI) [32]. We used the cumulative proportions of incidence or mortality rates of neonatal disorders and the cumulative proportion of nations ranked by SDI to construct Lorenz curves. The greater the deviation of the Lorenz curve from the diagonal line, the higher the degree of inequality. The CCI was calculated as: $\text{CCI} = \frac{2}{\mu} \text{cov}(h, r)$, where h refers to incidence rates or mortality rates, μ refers to mean (h), cov denotes covariance, and r represents the SDI ranking in the aggregate (in descending order), ranges from -1 to $+1$. A CCI of 0 signifies complete equal distribution with the Lorenz curve aligning with the equality line. A positive CCI (> 0) indicates a higher burden in wealthier (high-SDI) countries, and vice versa. All statistical analyses

were conducted in R (version 4.4.2) and two-tailed p values of less than 0.05 were considered to be statistically significant.

Results

Global burden and trends of neonatal disorders

In 2021, there were 27,062,768 new cases of neonatal disorders globally, of which 21,553,771 cases (79.64%) were due to neonatal preterm birth (Table 1). Between 2000 and 2021, the incidence rate of neonatal disorders declined from 23,112.05 per 100,000 live births in 2000 to 20,916.69 per 100,000 live births in 2021, with an EAPC of -0.63% (95% CI: -0.69 to -0.56) (Table 1 and Fig. 1). Since 2000, neonatal preterm birth has consistently had the highest incidence rate (16,658.81 per 100,000 live births in 2021, EAPC = -0.47%), followed by neonatal sepsis and other neonatal infections (2999.48 per 100,000 live births in 2021, EAPC = -1.43%), neonatal encephalopathy due to birth asphyxia and trauma (820.39 per 100,000 live births in 2021, EAPC = -0.80%), and hemolytic disease and other neonatal jaundice (438.01 per 100,000 live births in 2021, EAPC = -0.21%).

As shown in Table 2, in 2021, there were 1,831,535 deaths due to neonatal disorders globally, with a decrease of 31.12% compared to 2000. The main contributors were neonatal preterm birth, accounting for 40.39% (739,671 deaths), and neonatal encephalopathy due to birth asphyxia and trauma, accounting for 32.96% (603,606 deaths). Additionally, between 2000 and 2021, the global neonatal disorders mortality rate declined from 2048.34 per 100,000 live births to 1415.59 per 100,000 live births, with an EAPC of -1.76% (Table 2 and Fig. 1). Lorenz curves and concentrati also showed downward trends between 2000 and 2021 (all EAPCs and 95% CIs < 0). The mortality rates of neonatal preterm birth and neonatal encephalopathy due to birth asphyxia and trauma remained higher than the mortality rates of neonatal sepsis and other neonatal infections, hemolytic disease and other neonatal jaundice, and other neonatal disorders during the past two decades.

Regional burden and trends of neonatal disorders

In 2021, the highest burden of new cases for neonatal disorders was in South Asia (9,774,054), Western Sub-Saharan Africa (3,968,495), and Eastern Sub-Saharan Africa (3,201,823) (Table 1). South Asia had the highest incidence rate in 2021 (30,502.57 per 100,000 live births), followed by Southern Sub-Saharan Africa (24904.79 per 100,000 live births), Eastern Sub-Saharan Africa (23219.55 per 100,000 live births), and Western Sub-Saharan Africa (22246.13 per 100,000 live births). Between 2000 and 2021, Southern Latin America was among the seven regions with an increasing incidence

Table 1 Global and regional incident cases and incidence rates (per 100,000 live births) of neonatal disorders with changing trends, 2000–2021

Location	Incident case in 2000 (95% UI)	Incident case in 2021 (95% UI)	Relative change (%, 95 CI)	Incidence rate in 2000 (95% UI)	Incidence rate in 2021 (95% UI)	EAPC (%, 95 CI)
Global	30,001,427 (29674877 to 30440170)	27,062,768 (26800785 to 27342185)	-9.80% (-9.80 to -9.79)	23112.05 (22860.49 to 23450.04)	20916.69 (20714.21 to 21132.65)	-0.63% (-0.69 to -0.56)
SDI regions						
Low SDI	8,082,650 (7949069 to 8248566)	8,925,083 (8800000 to 9050568)	10.42% (10.41 to 10.43)	29246.82 (28763.46 to 29847.18)	24644.97 (24299.58 to 24991.48)	-0.88% (-0.93 to -0.84)
Low-middle SDI	11,771,060 (11595052 to 11994146)	9,574,083 (9431996 to 9725708)	-18.66% (-18.68 to -18.65)	29287.04 (28849.12 to 29842.08)	24355.28 (23993.83 to 24741.00)	-1.00% (-1.07 to -0.93)
Middle SDI	6,506,272 (6432041 to 6605875)	5,574,651 (5514146 to 5644344)	-14.32% (-14.34 to -14.30)	17919.24 (17714.79 to 18193.56)	17467.59 (17278.01 to 17685.97)	-0.26% (-0.33 to -0.19)
High-middle SDI	2,090,972 (2052103 to 2133114)	1,573,463 (1547917 to 1601803)	-24.75% (-24.80 to -24.7)	14692.54 (14419.42 to 14988.66)	13540.05 (13320.22 to 13783.92)	-0.52% (-0.60 to -0.44)
High SDI	1,529,966 (1510427 to 1550282)	1,395,013 (1375119 to 1416009)	-8.82% (-8.84 to -8.80)	13494.62 (13322.29 to 13673.82)	13650.39 (13455.73 to 13855.84)	-0.04% (-0.09 to 0.00)
GBD regions						
Andean Latin America	167,139 (159503 to 175650)	167,081 (159810 to 174279)	-0.03% (-0.03 to -0.03)	13766.85 (13137.89 to 14467.89)	13446.03 (12860.85 to 14025.28)	-0.04% (-0.10 to 0.02)
Australasia	34,567 (32907 to 36146)	40,642 (37785 to 43435)	17.57% (17.32 to 17.83)	11504.26 (10951.78 to 12029.82)	11356.36 (10558.00 to 12136.91)	-0.07% (-0.09 to -0.05)
Caribbean	178,022 (172278 to 184787)	175,414 (168401 to 181141)	-1.46% (-1.47 to -1.46)	20012.20 (19366.49 to 20772.71)	21250.28 (20400.78 to 21944.13)	0.27% (0.22 to 0.31)
Central Asia	219,496 (212186 to 227475)	277,474 (267360 to 288584)	26.41% (26.27 to 26.56)	14972.74 (14474.06 to 15517.02)	13384.60 (12896.70 to 13920.5)	-0.57% (-0.59 to -0.54)
Central Europe	209,954 (204456 to 216236)	165,762 (162332 to 169566)	-21.05% (-21.18 to -20.91)	17324.08 (16870.46 to 17842.45)	15965.26 (15634.98 to 16331.73)	-0.43% (-0.46 to -0.41)
Central Latin America	693,141 (678407 to 708366)	559,589 (547559 to 573039)	-19.27% (-19.34 to -19.20)	14109.34 (13809.42 to 14419.25)	14432.35 (14122.09 to 14779.24)	0.13% (0.11 to 0.16)
Central Sub-Saharan Africa	680,707 (649109 to 711978)	766,051 (732022 to 804546)	12.54% (12.50 to 12.58)	20503.21 (19551.47 to 21445.11)	17178.61 (16415.51 to 18041.84)	-0.85% (-0.91 to -0.80)
East Asia	1,733,498 (1700755 to 1766184)	995,451 (975543 to 1016032)	-42.58% (-42.68 to -42.47)	10691.57 (10489.62 to 10893.16)	8885.63 (8707.93 to 9069.34)	-0.89% (-0.97 to -0.80)
Eastern Europe	300,975 (288462 to 314075)	253,496 (246204 to 262054)	-15.78% (-15.86 to -15.69)	15692.14 (15039.75 to 16375.14)	14124.47 (13718.18 to 14601.31)	-0.57% (-0.63 to -0.52)
Eastern Sub-Saharan Africa	3,083,420 (3009014 to 3168492)	3,201,823 (3141786 to 3267127)	3.84% (3.83 to 3.85)	27358.05 (26697.87 to 28112.87)	23219.55 (22784.16 to 23693.14)	-0.82% (-0.84 to -0.79)
High-income Asia Pacific	174,768 (169427 to 179657)	114,404 (111465 to 117813)	-34.54% (-34.80 to -34.28)	9592.43 (9299.31 to 9860.81)	9781.94 (9530.65 to 10073.42)	0.04% (0.00 to 0.09)
High-income North America	664,164 (654015 to 674337)	581,904 (572766 to 591850)	-12.39% (-12.43 to -12.34)	15209.18 (14976.77 to 15442.13)	14494.35 (14266.73 to 14742.09)	-0.44% (-0.54 to -0.34)
North Africa and Middle East	2,472,728 (2411587 to 2530702)	2,462,558 (2397805 to 2525136)	-0.41% (-0.41 to -0.41)	21384.48 (20855.73 to 21885.85)	20267.98 (19735.03 to 20783.02)	-0.28% (-0.29 to -0.26)

Table 1 (continued)

Location	Incident case in 2000 (95% UI)	Incident case in 2021 (95% UI)	Relative change (% 95 CI)	Incidence rate in 2000 (95% UI)	Incidence rate in 2021 (95% UI)	EAPC (% 95 CI)
Oceania	56,943 (53522 to 60145)	87,947 (82162 to 93118)	54.45% (53.87 to 55.02)	19391.23 (18226.35 to 20481.85)	19476.03 (18194.98 to 20621.29)	0.04% (0.00 to 0.08)
South Asia	12,633,730 (12446592 to 12852318)	9,774,054 (9628748 to 9932007)	-22.64% (-22.65 to -22.62)	36289.24 (35751.70 to 36917.11)	30502.57 (30049.10 to 30995.50)	-0.96% (-1.03 to -0.89)
Southeast Asia	2,120,850 (2078834 to 2164311)	1,808,371 (1775925 to 1843685)	-14.73% (-14.76 to -14.70)	17819.87 (17466.84 to 18185.04)	16165.24 (15875.20 to 16480.91)	-0.52% (-0.55 to -0.49)
Southern Latin America	133,214 (126217 to 140003)	113,944 (107956 to 119864)	-14.47% (-14.58 to -14.35)	13420.55 (12715.66 to 14104.56)	14810.08 (14031.76 to 15579.57)	0.50% (0.47 to 0.53)
Southern Sub-Saharan Africa	432,911 (419613 to 450072)	407,640 (395160 to 419759)	-5.84% (-5.86 to -5.81)	26269.50 (25462.60 to 27310.88)	24904.79 (24142.34 to 25645.23)	-0.35% (-0.42 to -0.28)
Tropical Latin America	565,160 (553837 to 577562)	561,844 (550297 to 574165)	-0.59% (-0.59 to -0.58)	15333.38 (15026.17 to 15669.86)	16236.53 (15902.82 to 16592.56)	0.38% (0.30 to 0.45)
Western Europe	578,019 (565759 to 591490)	578,824 (563197 to 593245)	0.14% (0.14 to 0.14)	13331.86 (13049.10 to 13642.58)	14145.12 (13763.24 to 14497.54)	0.34% (0.30 to 0.38)
Western Sub-Saharan Africa	2,868,023 (2810378 to 2931579)	3,968,495 (3897001 to 4035764)	38.37% (38.31 to 38.43)	24402.65 (23912.18 to 24943.42)	22246.13 (21845.36 to 22623.22)	-0.50% (-0.56 to -0.44)
Categories of causes						
Neonatal preterm birth	23,095,467 (22927300 to 23268339)	21,553,771 (21378910 to 21730108)	-6.68% (-6.68 to -6.67)	17791.94 (17662.39 to 17925.11)	16658.81 (16523.66 to 16795.10)	-0.47% (-0.55 to -0.39)
Neonatal encephalopathy due to birth asphyxia and trauma	1,245,561 (1230678 to 1262518)	1,061,448 (1047815 to 1076766)	-14.78% (-14.82 to -14.74)	959.54 (948.07 to 972.60)	820.39 (809.85 to 832.23)	-0.80% (-0.82 to -0.77)
Neonatal sepsis and other neonatal infections	5,081,395 (4854033 to 5462140)	3,880,834 (3786667 to 3990588)	-23.63% (-23.66 to -23.60)	3914.53 (3739.38 to 4207.84)	2999.48 (2926.70 to 3084.31)	-1.43% (-1.58 to -1.28)
Hemolytic disease and other neonatal jaundice	579,004 (438112 to 763647)	566,715 (429507 to 746850)	-2.12% (-2.13 to -2.11)	446.04 (337.51 to 588.29)	438.01 (331.96 to 577.24)	-0.21% (-0.28 to -0.14)
Other neonatal disorders	/	/	/	/	/	/

CI confidence interval, EAPC estimated annual percentage change, GBD global burden of disease, SD/socio-demographic index; UI uncertain interval

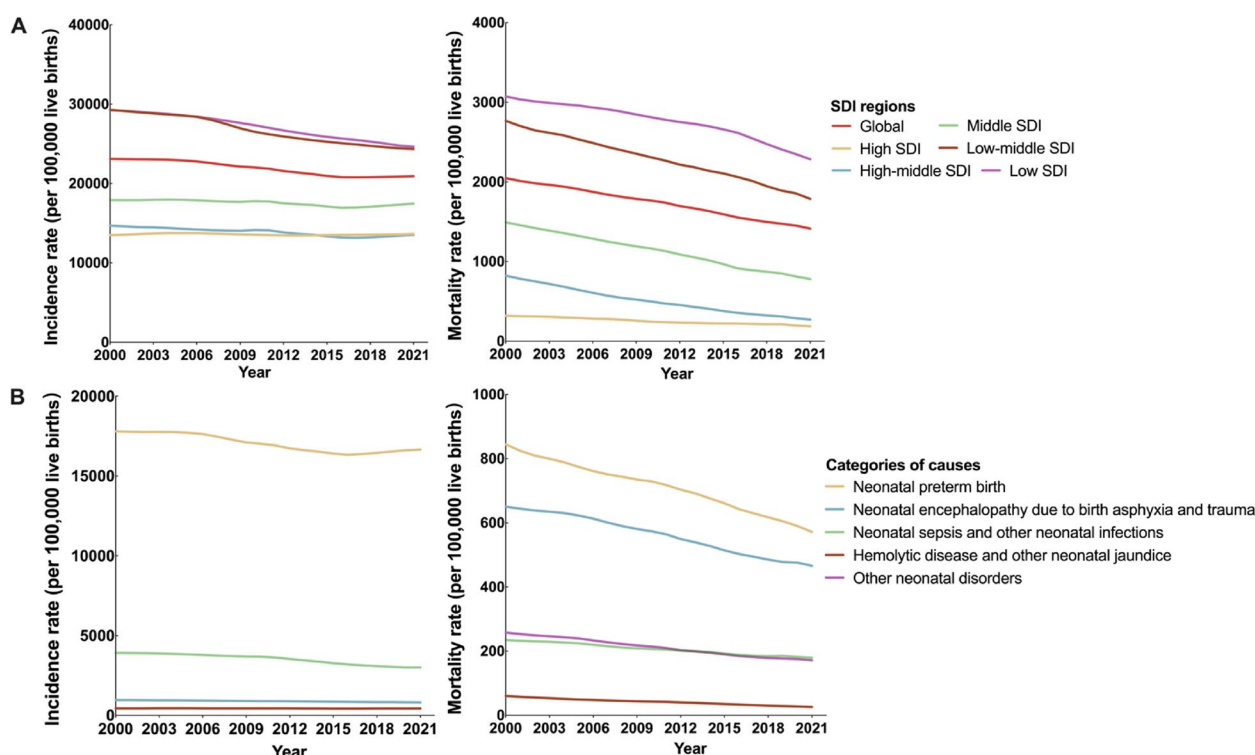


Fig. 1 Incidence and mortality rates of neonatal disorders by SDI **A** and cause **B** from 2000 to 2021. *SDI* Socio-demographic index

trend (EAPC = 0.50%, 95% CI: 0.47–0.53), whereas 13 regions experienced declines (all EAPCs and 95% CIs < 0), with the fastest decrease in South Asia (EAPC = −0.96%, 95% CI: −1.03 to −0.89).

Similarly, South Asia (636,120), Western Sub-Saharan Africa (481,305), and Eastern Sub-Saharan Africa (281,925) had the most deaths due to neonatal disorders in 2021 (Table 2). Sub-Saharan Africa (Western: 2698.04 per 100,000 live births; Eastern: 2044.52; Central: 1817.42; Southern: 1758.87) and South Asia (1985.18 per 100,000 live births) also had the highest neonatal disorders mortality rates in 2021. The mortality rates declined across all GBD regions from 2000 to 2021, with the fastest reductions in East Asia (EAPC = −6.55%, 95% CI: −6.69 to −6.41), Eastern Europe (EAPC = −4.84%, 95% CI: −5.24 to −4.43), and Central Europe (EAPC = −4.81%, 95% CI: −5.24 to −4.54).

Regarding the individual causes, the highest incidence rates in 2021 were as follows: neonatal preterm birth in South Asia (25,691.27 per 100,000 live births), neonatal encephalopathy due to birth asphyxia and trauma in Eastern Sub-Saharan Africa (1645.03 per 100,000 live births), neonatal sepsis and other neonatal infections in Southern Sub-Saharan Africa (7286.94 per 100,000 live births), and hemolytic disease and other neonatal jaundice in Andean Latin America (808.77 per 100,000 live births) (Table S2–S5). Between 2000 and 2021, the

incidence rates of neonatal preterm birth increased in 11 of the 21 GBD regions (all EAPCs and 95% CIs > 0) and decreased in 10 regions (all EAPCs and 95% CIs < 0). The incidence rates of neonatal encephalopathy due to birth asphyxia and trauma declined across almost all GBD regions, except for High-income Asia Pacific. The incidence rates of neonatal sepsis and other neonatal infections increased only in Andean Latin America, while it declined in 18 regions. The incidence rates of hemolytic disease and other neonatal jaundice increased in four regions, while it decreased in 15 regions. In terms of mortality rates in 2021, Western Sub-Saharan Africa had the highest rates for neonatal preterm birth (1023.29 per 100,000 live births), neonatal encephalopathy due to birth asphyxia and trauma (1053.78 per 100,000 live births), and neonatal sepsis and other neonatal infections (406.35 per 100,000 live births), while mortality for other neonatal disorders was mainly concentrated in Sub-Saharan Africa and South Asia (Table S6–S10). Between 2000 and 2021, the mortality rates for all individual causes showed downward trends across almost all GBD regions (EAPCs and 95% CIs < 0), with very few exceptions. Overall, the highest burden of both incidence and mortality from neonatal disorders remained concentrated in South Asia and Sub-Saharan Africa, with consistent downward trends observed across most regions over the past two decades.

Table 2 Global and regional deaths and mortality rates (per 100,000 live births) of neonatal disorders with changing trends, 2000–2021

Location	Death in 2000 (95% UI)	Death in 2021 (95% UI)	Relative change (%, 95 CI)	Mortality rate in 2000 (95% UI)	Mortality rate in 2021 (95% UI)	EAPC (%, 95 CI)
Global	2,658,922 (2506647 to 2822892)	1,831,535 (1570938 to 2121829)	-31.12% (-31.18 to -31.06)	2048.34 (1931.03 to 2174.66)	1415.59 (1214.17 to 1639.95)	-1.76% (-1.82 to -1.69)
SDI regions						
Low SDI	849,554 (795346 to 906961)	827,348 (688075 to 988831)	-2.61% (-2.62 to -2.61)	3074.09 (2877.94 to 3281.81)	2284.57 (1899.99 to 2730.48)	-1.29% (-1.43 to -1.14)
Low-middle SDI	1,112,368 (1033045 to 1192106)	702,365 (602557 to 822232)	-36.86% (-36.97 to -36.75)	2767.63 (2570.27 to 2966.03)	1786.73 (1532.83 to 2091.66)	-1.96% (-2.04 to -1.89)
Middle SDI	542,116 (508114 to 578283)	249,633 (214218 to 291246)	-53.95% (-54.21 to -53.7)	1493.07 (1399.42 to 1592.68)	782.20 (671.23 to 912.59)	-3.05% (-3.18 to -2.92)
High-middle SDI	117,143 (109021 to 126589)	31,849 (27927 to 35729)	-72.81% (-73.71 to -71.91)	823.13 (766.05 to 889.5)	274.07 (240.32 to 307.46)	-5.12% (-5.19 to -5.04)
High SDI	36,357 (35092 to 37814)	19,276 (17286 to 21077)	-46.98% (-47.80 to -46.16)	320.67 (309.51 to 333.52)	188.62 (169.14 to 206.24)	-2.44% (-2.59 to -2.29)
GBD regions						
Andean Latin America	16,272 (14737 to 17795)	7390 (5828 to 9129)	-54.58% (-56.09 to -53.08)	1340.29 (1213.87 to 1465.74)	594.72 (469.04 to 734.63)	-3.40% (-3.63 to -3.16)
Australasia	780 (748 to 815)	528 (446 to 629)	-32.31% (-35.88 to -28.74)	259.58 (248.95 to 271.17)	147.59 (124.53 to 175.69)	-2.31% (-2.57 to -2.06)
Caribbean	12,299 (11224 to 13359)	10,549 (8534 to 12806)	-14.23% (-14.60 to -13.86)	1382.54 (1261.71 to 1501.71)	1277.90 (1033.81 to 1551.42)	-0.28% (-0.38 to -0.19)
Central Asia	17,956 (16542 to 19540)	15,458 (13355 to 18088)	-13.91% (-14.21 to -13.61)	1224.84 (1128.42 to 1332.89)	745.66 (644.20 to 872.49)	-2.66% (-3.03 to -2.29)
Central Europe	6515 (6260 to 6778)	2013 (1712 to 2348)	-69.1% (-72.56 to -65.65)	537.58 (516.58 to 559.29)	193.83 (164.89 to 226.13)	-4.81% (-5.08 to -4.54)
Central Latin America	54,735 (50470 to 59564)	23,796 (18809 to 29962)	-56.53% (-57.39 to -55.66)	1114.16 (1027.35 to 1212.46)	613.72 (485.11 to 772.74)	-2.43% (-2.64 to -2.23)
Central Sub-Saharan Africa	77,168 (68514 to 86624)	81,045 (63220 to 96901)	5.02% (4.97 to 5.07)	2324.35 (2063.67 to 2609.14)	1817.42 (1417.69 to 2172.99)	-1.03% (-1.50 to -0.55)
East Asia	169,701 (152499 to 187671)	29,005 (24894 to 34144)	-82.91% (-83.94 to -81.88)	1046.65 (940.56 to 1157.48)	258.90 (222.21 to 304.78)	-6.55% (-6.69 to -6.41)
Eastern Europe	12,322 (11861 to 12826)	3808 (3404 to 4219)	-69.10% (-71.61 to -66.59)	642.45 (618.42 to 668.72)	212.15 (189.66 to 235.11)	-4.84% (-5.24 to -4.43)
Eastern Sub-Saharan Africa	335,035 (307850 to 362948)	281,925 (226651 to 347632)	-15.85% (-15.93 to -15.77)	2972.65 (2731.44 to 3220.30)	2044.52 (1643.67 to 2521.02)	-1.69% (-1.84 to -1.55)
High-income Asia Pacific	2763 (2644 to 2911)	758 (684 to 836)	-72.57% (-78.40 to -66.73)	151.67 (145.10 to 159.77)	64.80 (58.49 to 71.47)	-3.70% (-3.93 to -3.48)
High-income North America	15,746 (15377 to 16113)	10,475 (9353 to 11641)	-33.48% (-34.30 to -32.65)	360.57 (352.13 to 368.98)	260.91 (232.96 to 289.96)	-1.43% (-1.63 to -1.23)
North Africa and Middle East	195,786 (175127 to 219815)	86,799 (74344 to 100605)	-55.67% (-56.11 to -55.22)	1693.18 (1514.52 to 1900.99)	714.4 (611.89 to 828.02)	-4.06% (-4.21 to -3.92)

Table 2 (continued)

Location	Death in 2000 (95% UI)	Death in 2021 (95% UI)	Relative change (%, 95 CI)	Mortality rate in 2000 (95% UI)	Mortality rate in 2021 (95% UI)	EAPC (%, 95 CI)
Oceania	3074 (2604 to 3626)	4555 (3645 to 5561)	48.18% (45.97 to 50.38)	1046.74 (886.62 to 1234.91)	1008.70 (807.26 to 1231.43)	-0.18% (-0.34 to -0.02)
South Asia	1,055,283 (960,787 to 1,158,715)	636,120 (538,400 to 754,316)	-39.72% (-39.84 to -39.60)	3031.20 (2759.77 to 3328.30)	1985.18 (1680.22 to 2354.05)	-1.84% (-1.96 to -1.72)
Southeast Asia	174,660 (163,016 to 188,096)	97,282 (81,801 to 113,664)	-44.30% (-44.65 to -43.95)	1467.53 (1369.70 to 1580.42)	869.61 (731.23 to 1016.05)	-2.47% (-2.53 to -2.41)
Southern Latin America	7394 (7148 to 7640)	2938 (2308 to 3668)	-60.27% (-62.84 to -57.69)	744.91 (720.13 to 769.68)	381.84 (299.97 to 476.77)	-3.02% (-3.25 to -2.79)
Southern Sub-Saharan Africa	30,718 (28300 to 33168)	28,789 (23911 to 34813)	-6.28% (-6.38 to -6.18)	1864.00 (1717.26 to 2012.68)	1758.87 (1460.84 to 2126.92)	-0.60% (-0.76 to -0.43)
Tropical Latin America	55,078 (48,215 to 62,487)	20,731 (16,712 to 25,743)	-62.36% (-63.36 to -61.36)	1494.33 (1308.12 to 1695.35)	599.11 (482.96 to 743.95)	-4.23% (-4.45 to -4.02)
Western Europe	10,300 (10,067 to 10,524)	6267 (5503 to 7000)	-39.16% (-40.38 to -37.93)	237.57 (232.20 to 242.74)	153.14 (134.47 to 171.06)	-1.94% (-2.09 to -1.78)
Western Sub-Saharan Africa	405,337 (379,015 to 433,578)	481,305 (407,468 to 561,656)	18.74% (18.66 to 18.82)	3448.82 (3224.86 to 3689.11)	2698.04 (2284.14 to 3148.47)	-1.01% (-1.14 to -0.88)
Categories of causes						
Neonatal preterm birth	1,096,580 (1,005,656 to 1,193,709)	739,671 (622,746 to 877,043)	-32.55% (-32.64 to -32.45)	844.77 (774.72 to 919.59)	571.69 (481.32 to 677.86)	-1.73% (-1.82 to -1.63)
Neonatal encephalopathy due to birth asphyxia and trauma	844,622 (783,450 to 938,896)	603,606 (511,191 to 725,271)	-28.54% (-28.63 to -28.44)	650.67 (603.54 to 723.29)	466.52 (395.10 to 560.56)	-1.70% (-1.78 to -1.61)
Neonatal sepsis and other neonatal infections	304,594 (273,807 to 337,556)	232,656 (197,144 to 272,485)	-23.62% (-23.75 to -23.49)	234.65 (210.93 to 260.04)	179.82 (152.37 to 210.60)	-1.34% (-1.39 to -1.29)
Hemolytic disease and other neonatal jaundice	78,124 (66,667 to 97,058)	33,661 (25,908 to 43,078)	-56.91% (-57.64 to -56.19)	60.18 (51.36 to 74.77)	26.02 (20.02 to 33.30)	-3.71% (-3.89 to -3.52)
Other neonatal disorders	335,002 (238,812 to 397,875)	221,941 (159,906 to 268,608)	-33.75% (-33.93 to -33.57)	258.07 (183.97 to 306.51)	171.54 (123.59 to 207.61)	-2.04% (-2.11 to -1.98)

CI confidence interval, EAPC estimated annual percentage change, GBD global burden of disease, SD/ socio-demographic index, UI uncertain interval

National burden and trends of neonatal disorders

In 2021, among 204 countries and territories, India, Nigeria, and Pakistan reported the highest numbers of new cases and deaths from neonatal disorders (Table S11). Compared to 2000, the number of new cases increased in 76 nations, while the number of deaths rose in 22 nations in 2021. Similarly, all individual causes ranked among the highest in terms of both incidence and deaths

in India, Nigeria, and Pakistan. Among the 204 countries and territories, Mauritania, India, Botswana, and Yemen had the highest incidence rates of neonatal disorders, all exceeding 33,000 cases per 100,000 live births (Table S11 and Fig. 2). From 2000 to 2021, the incidence rates of neonatal disorders increased in 44 countries, particularly in Greece (EAPC = 2.22%, 95% CI: 1.80 to 2.65), Austria (EAPC = 1.51%, 95% CI: 1.22 to 1.79), and Portugal

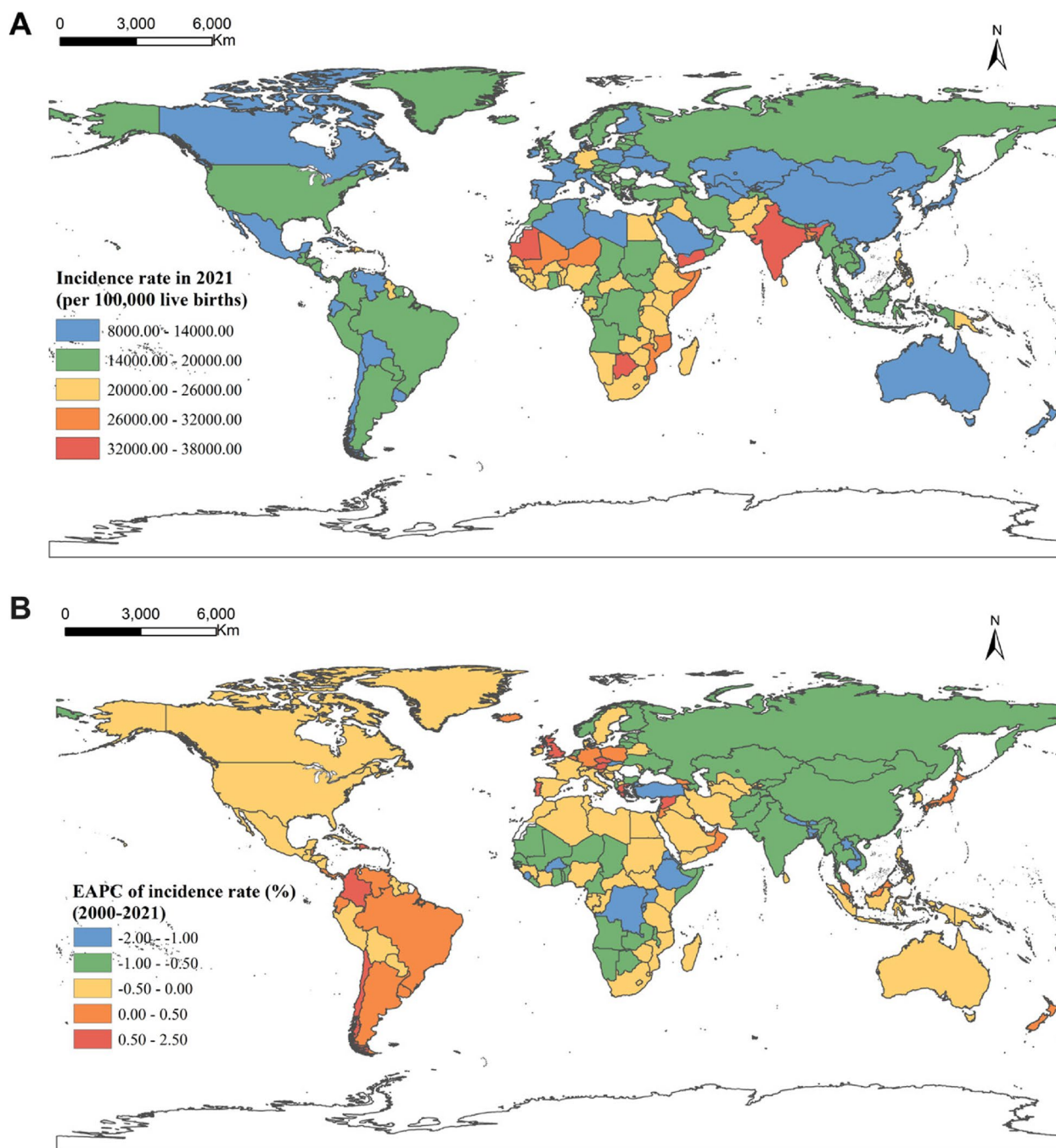


Fig. 2 Incidence rates of neonatal disorders in 2021 and EAPCs across 204 countries and territories between 2000 and 2021. EAPC Estimated annual percentage change

(EAPC = 1.25%, 95% CI: 1.04 to 1.46). In contrast, 142 countries saw a decrease, especially Cambodia (EAPC = -1.89%, 95% CI: -2.13 to -1.65), Nepal (EAPC = -1.81%, 95% CI: -1.88 to -1.74), and Rwanda (EAPC = -1.74%, 95% CI: -1.97 to -1.51). In addition, Mali, South Sudan, Sierra Leone, and Pakistan had the highest neonatal mortality rates, all exceeding 3300 deaths per 100,000 live births (Table S12). Between 2000 and 2021, neonatal mortality increased in nine countries, particularly in

Dominica (EAPC = 2.57%, 95% CI: 2.09 to 3.04), Northern Mariana Islands (EAPC = 1.49%, 95% CI: 0.87 to 2.11), and Venezuela (EAPC = 1.11%, 95% CI: 0.27 to 1.95) (Table S12 and Fig. 3). Conversely, 187 countries showed a decline, with Saudi Arabia (EAPC = -8.50%, 95% CI: -8.70 to -8.31) and Montenegro (EAPC = -8.48%, 95% CI: -8.93 to -8.03) experiencing the fastest drop.

As for the individual causes, the highest incidence rates of neonatal preterm birth, neonatal encephalopathy due

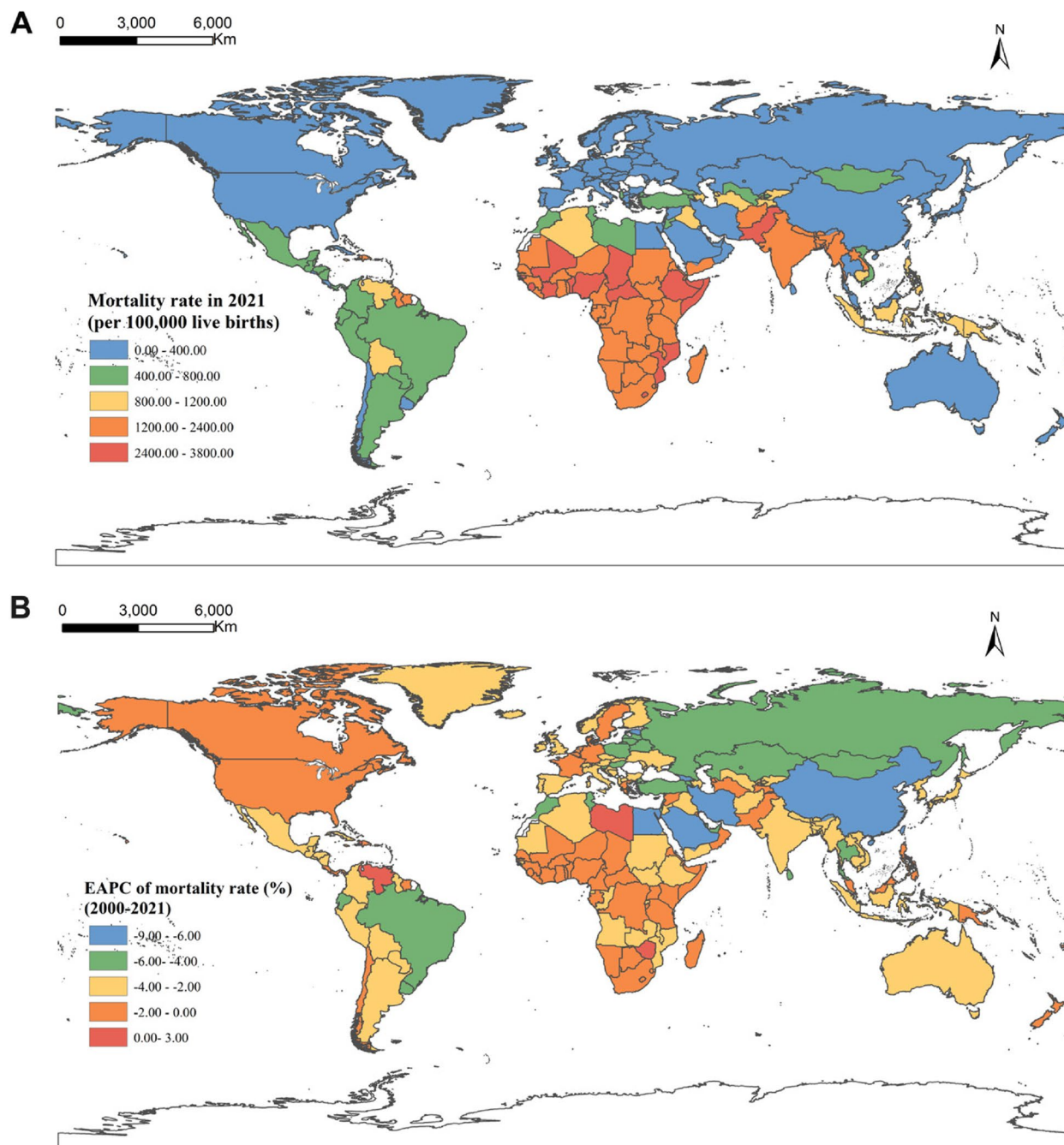


Fig. 3 Mortality rates of neonatal disorders in 2021 and EAPCs across 204 countries and territories between 2000 and 2021. EAPC Estimated annual percentage change

to birth asphyxia and trauma, neonatal sepsis and other neonatal infections, and hemolytic disease and other neonatal jaundice were found in Mauritania (30,916.43 per 100,000 live births), Somalia (2694.44 per 100,000 live births), Romania (9776.46 per 100,000 live births), and Albania (1471.63 per 100,000 live births), respectively. Meanwhile, the highest mortality rates of neonatal preterm birth, neonatal encephalopathy due to birth asphyxia and trauma, neonatal sepsis and other neonatal infections, and hemolytic disease and other neonatal jaundice were recorded in Mali (1872.34 per 100,000 live births), Pakistan (1552.60 per 100,000 live births), Chad (517.38 per 100,000 live births), and Pakistan (139.77 per 100,000 live births). Despite overall global declines, substantial national disparities persist, with several low-income countries continuing to bear a disproportionately high burden of neonatal morbidity and mortality (Table S13–17).

Global trends in neonatal disorders before and during the COVID-19 pandemic

Globally, the incidence rates of neonatal disorders shifted from a declining trend during the pre-pandemic period (2000–2019: EAPC = -0.68% , 95% CI: -0.74 to -0.62) to an increasing trend during the pandemic period (2019–2021: EAPC = 0.17% , 95% CI: 0.13 to 0.22) (Table 3 and Fig. 4). Similar changes were observed in regions with middle SDI, high-middle SDI, and high SDI. In contrast, low and low-middle SDI regions continued to experience declining incidence rates, though the rate of decline slowed during the pandemic. The incidence rates of neonatal disorders in most GBD regions slowed down in their declining trends or even reversed to an increasing trend during the COVID-19 pandemic, except for Central Latin America, Central Sub-Saharan Africa, Eastern Sub-Saharan Africa, and Western Sub-Saharan Africa. In contrast, the increasing trends in High-income Asia Pacific, Oceania, Southern Latin America, Tropical Latin America, and Western Europe slowed down or reversed into a declining trend during the pandemic. Additionally, the global incidence rates of neonatal preterm birth shifted from a declining trend in the pre-pandemic period to an increasing trend during the pandemic period, while the neonatal encephalopathy due to birth asphyxia and trauma, as well as neonatal sepsis and other neonatal infections, showed a slowed decline during the pandemic (Table 3 and Table S18–S21). National changes in incidence and mortality rates for neonatal disorders before and during the COVID-19 pandemic (2000–2019 vs. 2019–2021) are presented in Figure S1–S2.

Globally, the mortality rate of neonatal disorders continued to decline during the pandemic, with a faster change in the rate of decline (EAPC: -1.73% vs. -2.04%) (Table 3 and Fig. 4). All SDI regions and the vast majority

of GBD regions (except for Oceania and Southern Sub-Saharan Africa) accelerated the downward trend during the pandemic period. The decreases in global mortality rates for neonatal preterm birth, neonatal sepsis and other neonatal infections, and hemolytic disease and other neonatal jaundice were more significant in 2019–2021, while the decline in mortality rates for the other three causes slowed (Table 3 and Table S18–S22).

Cross-country health inequality and concentration indices contributed by SDI

As shown in Table 1, in 2021, the incidence rates and mortality rates of neonatal disorders and all individual causes were decreased with increasing SDI, and all SDI regions showed downward trends between 2000 and 2021 (all EAPCs and 95% CIs < 0). Figure 5 and Figures S3–S4 present the Lorenz curves and concentration indices for the cumulative distribution of global incidence and mortality rates of neonatal disorders, as well as all individual causes, across different SDI countries from 2000 to 2021. Notably, all Lorenz curves were above the line of equality and statistically significant, indicating that from 2000 to 2021, the global burden of neonatal disorders was concentrated in lower SDI countries, particularly in terms of mortality rates. Moreover, all CCIs remained around -0.5 , with only slight variations, suggesting that inequalities in the burden of neonatal disorders related to SDI have persisted between 2000 and 2021.

Discussion

To our knowledge, this is the first comprehensive study to quantify the global burden, long-term trends, and inequalities of neonatal disorders in the 21st century. Our results found that, in 2021, there were a total of 27,062,768 new cases and 1,831,535 deaths due to neonatal disorders. From 2000 to 2021, the global incidence and mortality rates of neonatal disorders exhibited an overall downward trend, with neonatal preterm birth remaining the leading contributor. South Asia and sub-Saharan Africa were identified as high-burden regions. During the same period, seven GBD regions experienced an increasing incidence trend, while mortality rates declined across all GBD regions. The incidence rate of neonatal disorders declined before the COVID-19 pandemic but began to rise during the pandemic, whereas the mortality rate continued to decline at a faster pace than the pre-pandemic period. Additionally, neonatal disorders disproportionately affected countries with lower SDI levels, particularly in terms of mortality. This study offers data-driven insights to guide governments and international organizations in enhancing neonatal health interventions and mitigating global and regional health disparities.

Global progress in maternal and neonatal care over the past two decades can be attributed to both economic

Table 3 Global and regional trends in incidence and mortality rates for neonatal disorders before and during the COVID-19 pandemic (2000–2019 vs. 2019–2021)

Location	Incidence		Death	
	EAPC 00–19 (%, 95 CI)	EAPC 19–21 (%, 95 CI)	EAPC 00–19 (%, 95 CI)	EAPC 19–21 (%, 95 CI)
Global	−0.68% (−0.74 to −0.62)	0.17% (0.13 to 0.22)	−1.73% (−1.80 to −1.66)	−2.04% (−2.55 to −1.52)
SDI regions				
Low SDI	−0.88% (−0.93 to −0.82)	−0.81% (−1.05 to −0.57)	−1.15% (−1.26 to −1.03)	−2.65% (−2.80 to −2.5)
Low-middle SDI	−1.04% (−1.11 to −0.96)	−0.48% (−0.65 to −0.32)	−1.89% (−1.95 to −1.83)	−2.84% (−3.88 to −1.78)
Middle SDI	−0.32% (−0.39 to −0.25)	0.71% (0.62 to 0.80)	−2.98% (−3.13 to −2.83)	−4.21% (−4.58 to −3.84)
High-middle SDI	−0.59% (−0.66 to −0.52)	0.82% (0.80 to 0.84)	−5.07% (−5.16 to −4.99)	−6.26% (−7.71 to −4.79)
High SDI	−0.06% (−0.12 to −0.01)	0.25% (0.17 to 0.33)	−2.39% (−2.56 to −2.21)	−5.99% (−7.19 to −4.77)
GBD regions				
Andean Latin America	−0.05% (−0.12 to 0.02)	0.05% (−0.08 to 0.18)	−3.14% (−3.30 to −2.99)	−10.62% (−13.96 to −7.16)
Australasia	−0.07% (−0.10 to −0.04)	−0.02% (−0.12 to 0.07)	−2.2% (−2.45 to −1.95)	−10.67% (−14.02 to −7.18)
Caribbean	0.28% (0.22 to 0.33)	0.14% (0.06 to 0.22)	−0.21% (−0.30 to −0.11)	−0.93% (−2.32 to 0.49)
Central Asia	−0.59% (−0.61 to −0.56)	−0.36% (−0.88 to 0.16)	−2.52% (−2.95 to −2.09)	−3.74% (−5.87 to −1.55)
Central Europe	−0.46% (−0.48 to −0.44)	−0.03% (−0.09 to 0.02)	−4.87% (−5.19 to −4.54)	−6.37% (−8.37 to −4.32)
Central Latin America	0.15% (0.12 to 0.17)	−0.01% (−0.10 to 0.08)	−2.33% (−2.56 to −2.11)	−6.80% (−8.46 to −5.1)
Central Sub-Saharan Africa	−0.81% (−0.87 to −0.75)	−1.12% (−1.41 to −0.83)	−0.64% (−1.10 to −0.19)	−4.67% (−4.81 to −4.52)
East Asia	−0.96% (−1.04 to −0.88)	−0.16% (−0.3 to −0.03)	−6.49% (−6.65 to −6.33)	−8.10% (−8.47 to −7.73)
Eastern Europe	−0.58% (−0.64 to −0.51)	0.03% (−0.13 to 0.19)	−4.60% (−5.03 to −4.16)	−5.37% (−12.66 to 2.54)
Eastern Sub-Saharan Africa	−0.81% (−0.84 to −0.79)	−0.83% (−1.10 to −0.57)	−1.56% (−1.69 to −1.43)	−2.90% (−3.13 to −2.67)
High-income Asia Pacific	0.06% (0.01 to 0.11)	0.08% (−0.06 to 0.21)	−3.62% (−3.86 to −3.37)	−10.99% (−11.79 to −10.18)
High-income North America	−0.49% (−0.60 to −0.38)	0.23% (0.15 to 0.31)	−1.27% (−1.46 to −1.08)	−5.93% (−5.95 to −5.91)
North Africa and Middle East	−0.28% (−0.29 to −0.27)	−0.08% (−0.46 to 0.29)	−3.95% (−4.09 to −3.81)	−6.72% (−8.35 to −5.05)
Oceania	0.05% (0.00 to 0.1)	−0.42% (−0.46 to −0.37)	−0.26% (−0.45 to −0.08)	−0.16% (−0.39 to 0.07)
South Asia	−1.00% (−1.08 to −0.92)	−0.45% (−0.59 to −0.31)	−1.75% (−1.86 to −1.63)	−2.51% (−4.43 to −0.56)
Southeast Asia	−0.54% (−0.57 to −0.51)	−0.24% (−0.41 to −0.07)	−2.4% (−2.44 to −2.37)	−3.64% (−5.7 to −1.53)
Southern Latin America	0.53% (0.50 to 0.55)	0.08% (−0.01 to 0.18)	−3.03% (−3.30 to −2.75)	−7.45% (−7.48 to −7.42)
Southern Sub-Saharan Africa	−0.33% (−0.41 to −0.24)	−0.29% (−0.59 to 0.01)	−0.75% (−0.89 to −0.61)	3.85% (−1.33 to 9.3)
Tropical Latin America	0.41% (0.32 to 0.50)	0.30% (0.20 to 0.39)	−4.05% (−4.25 to −3.85)	−5.30% (−5.41 to −5.2)
Western Europe	0.36% (0.32 to 0.40)	0.09% (0.02 to 0.16)	−1.89% (−2.07 to −1.71)	−4.91% (−8.02 to −1.69)

Table 3 (continued)

Location	Incidence		Death	
	EAPC 00–19 (%, 95 CI)	EAPC 19–21 (%, 95 CI)	EAPC 00–19 (%, 95 CI)	EAPC 19–21 (%, 95 CI)
Western Sub-Saharan Africa	−0.47% (−0.54 to −0.41)	−0.66% (−0.94 to −0.38)	−0.87% (−0.95 to −0.79)	−2.82% (−3.21 to −2.43)
Categories of causes				
Neonatal preterm birth	−0.54% (−0.61 to −0.47)	0.39% (0.30 to 0.48)	−1.65% (−1.74 to −1.56)	−2.80% (−3.19 to −2.41)
Neonatal encephalopathy due to birth asphyxia and trauma	−0.80% (−0.83 to −0.77)	−0.64% (−0.68 to −0.61)	−1.69% (−1.80 to −1.59)	−1.23% (−2.11 to −0.34)
Neonatal sepsis and other neonatal infections	−1.39% (−1.57 to −1.21)	−0.76% (−1.48 to −0.04)	−1.36% (−1.42 to −1.3)	−1.55% (−1.79 to −1.30)
Hemolytic disease and other neonatal jaundice	−0.25% (−0.32 to −0.17)	0.15% (−0.28 to 0.59)	−3.55% (−3.72 to −3.38)	−5.35% (−5.59 to −5.11)
Other neonatal disorders	/	/	−2.08% (−2.15 to −2.02)	−1.60% (−2.43 to −0.76)

CI confidence interval; COVID-19 coronavirus disease 2019, EAPC estimated annual percentage change, GBD global burden of disease, SDI socio-demographic index

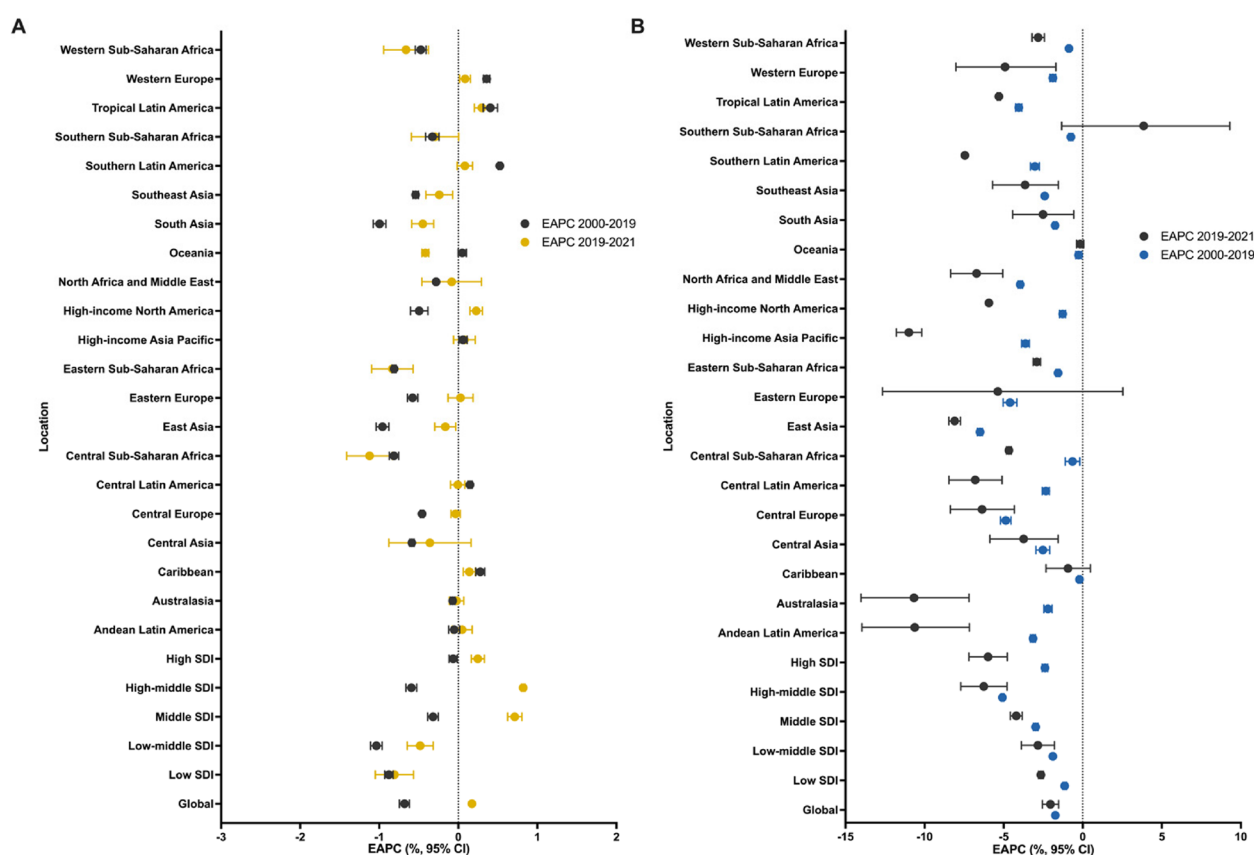


Fig. 4 Global and regional EAPCs in incidence **A** and mortality rates **B** for neonatal disorders before and during the COVID-19 pandemic (2000–2019 vs. 2019–2021). CI confidence interval; COVID-19 coronavirus disease 2019; EAPC Estimated annual percentage change; SDI socio-demographic index

development and strategic health policies. A major driver of this progress is Every Newborn Action Plan (ENAP), launched in 2014, which aims to end preventable newborn and maternal deaths by improving care around the time of birth [33, 34]. The success of ENAP and similar initiatives has contributed to substantial reductions in

neonatal mortality and morbidity, particularly in low-resource regions with a high burden [35–37]. However, South Asia and sub-Saharan Africa continue to bear a disproportionately high burden of morbidity (e.g., Mauritania, India, and Botswana) and mortality (e.g., Mali, South Sudan, Sierra Leone, and Pakistan) [10, 38, 39].

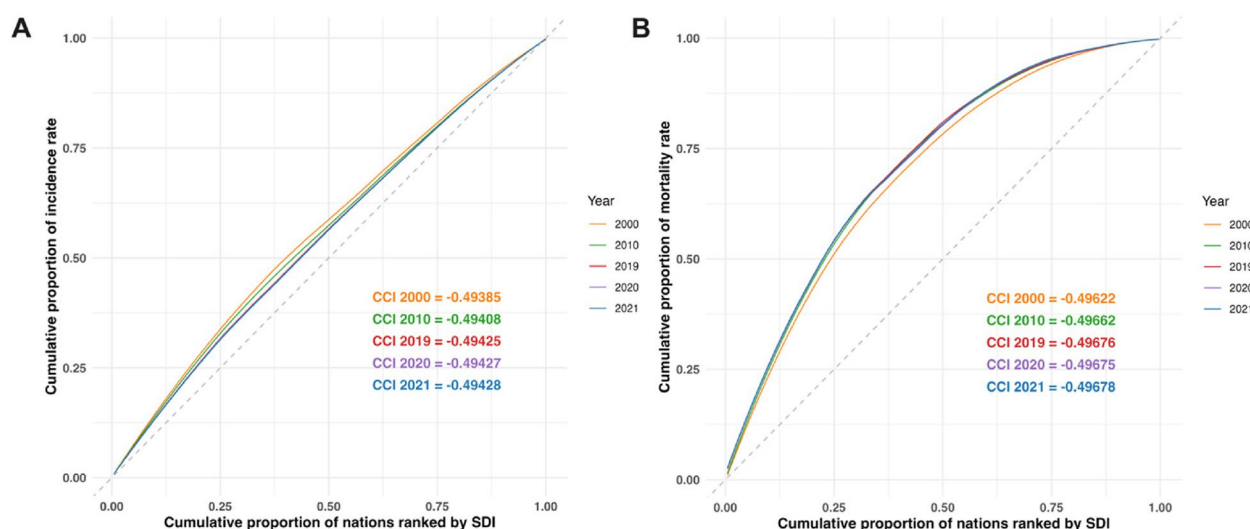


Fig. 5 Lorentz curves and concentration indices for the incidence and mortality rates of neonatal disorders between 2000 and 2021. CCI concentration index

According to global health reports from UNICEF and WHO, healthcare coverage in South Asia and sub-Saharan Africa remains limited, particularly in primary care and emergency obstetric services [40, 41]. Marchant et al. assessed maternal and newborn healthcare coverage and equity in rural Nigeria, Ethiopia, and India in both 2012 and 2015, finding that while community interventions had improved coverage, significant inequalities remained [42]. A previous study found that fewer than one in four eligible women in India received quality antenatal care, with most services concentrated in high-income areas, while rural areas had coverage rates below 10% [43]. Moreover, poor environmental sanitation, including inadequate hygiene facilities and unclean delivery conditions, further increases the risk of adverse outcomes, such as maternal and neonatal sepsis [44, 45]. A significant proportion of births in sub-Saharan Africa occurred without the assistance of skilled birth attendants, compounded by a substantial lack of high-quality prenatal and perinatal care [46, 47]. Studies have highlighted the strong association between malnutrition (maternal and neonatal) and neonatal complications [48–50]. Additionally, malnutrition in neonates may also compromise the immune system, increasing neonatal susceptibility to infections [50–52]. Low maternal education, maternal unemployment, and a lower household wealth index are among the most common socioeconomic factors linked to adverse birth outcomes and infant malnutrition [53, 54]. In regions such as North Africa, the Middle East, and Eastern Europe, ongoing conflicts and the resulting displacement of populations have profoundly impacted maternal and child health [55–57].

It is noteworthy that some countries with high and high-middle SDI, such as Greece, Austria, and Portugal,

experienced an upward trend in the incidence of neonatal disorders from 2000 to 2021. This pattern may reflect underlying shortcomings in the delivery of maternal and newborn care services, as well as inconsistencies in the implementation of relevant public health policies [58]. Another likely main contributor to the increasing burden of neonatal conditions may be the rising incidence of preterm births. Previous studies have highlighted a steady increase in preterm birth rates across high-income countries over the past two decades [59–63]. Several factors may be associated with this shift, including a higher frequency of multiple pregnancies, a growing proportion of births among individuals over the age of 34, expanded use of assisted reproductive technologies, and evolving clinical practices—such as the increased use of elective cesarean Sects [60, 64–66]. While these factors offer partial explanations, they may not fully account for the observed trend.

During the COVID-19 pandemic, the long-term trends in the global burden of neonatal disorders were affected in complex ways [67]. In our study, the incidence of neonatal diseases was primarily attributed to preterm birth. A noticeable upward trend in the global preterm birth rate since around 2016 has also been repeatedly reported in other studies [58, 60]. Disruptions in maternal and neonatal healthcare services, along with reduced access to prenatal care during the pandemic, may have contributed to the increase in incidence or the slower pace of decline [67–69]. Moreover, worsening maternal mental health issues and increased socioeconomic stress may have further elevated the incidence of preterm birth and other neonatal disorders [67, 70, 71]. The reallocation of medical resources during the pandemic likely led to a reduction in routine prenatal check-ups, particularly in

low-SDI regions, may have increased the risk of neonatal disorders [67–69]. Reduced prenatal follow-up could have delayed the identification of maternal-fetal blood type incompatibility, leading to inadequate early intervention for hemolytic disease [68]. Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection among pregnancy women could also contributed to fetal distress and preterm birth through inflammation-mediated placental dysfunction or direct effects on fetal growth and development [72]. Additionally, disruptions in the antibiotic supply chain could also affected the management of neonatal sepsis [73]. Interestingly, the mortality rate of neonatal disorders continued to decline globally, with certain regions even exhibiting an accelerated decrease. This trend likely reflects the effectiveness of government and healthcare system efforts to prioritize reducing neonatal and infant mortality through targeted resource allocation and policy interventions [20]. Advances in neonatal care technology and the widespread establishment of NICUs have significantly improved survival rates of high-risk neonates, such as preterm infants, those with severe jaundice, and infections [74, 75]. However, it is important to consider that some of the observed declines may be partly due to underreporting or misclassification of neonatal deaths, particularly in regions with incomplete vital registration systems. Moreover, during the COVID-19 pandemic, some neonatal deaths may have been directly attributed to SARS-CoV-2 infection rather than the specific neonatal conditions examined in this study, potentially leading to an underestimation of mortality related to those conditions [76]. Therefore, interpreting these mortality trends requires cautious consideration of both improvements in neonatal care and potential data limitations.

Over the past several decades, inequalities in newborn and child health have remained pressing social concerns [10]. Chen et al. found that in 2019, the age-standardized incidence and mortality rates of neonatal disorders across GBD regions were negatively correlated with SDI, which aligns with our findings [77]. Furthermore, Peng et al. noted that from 1990 to 2019, socioeconomic inequalities in neonatal disorders, still mainly affecting poorer populations, did not exhibit significant overall changes [10]. Resource-limited regions continue to face growing challenges in their healthcare systems due to the high burden of neonatal disorders [78]. Developed countries have seen a steady decline in the burden of neonatal disorders, largely driven by their well-developed healthcare systems and broad health service coverage [79–82]. Low-income countries are often characterized by rapid population growth, maternal malnutrition, high infection rates, inadequate prenatal screening, and limited healthcare resources [80, 81]. These factors not only contribute to elevated neonatal incidence and mortality rates but also

place an increasing strain on healthcare systems, hindering economic development and perpetuating a vicious cycle of poverty and poor health outcomes [83]. However, it is essential to emphasize that the persistence or worsening of pro-poor inequality may stem from multiple factors, such as healthcare resource allocation, policy implementation, cultural behaviors, and environmental conditions, rather than socioeconomic status alone [84, 85].

To effectively alleviate the burden of neonatal disorders and promote global health equity, priority should be given to increasing investment in maternal and neonatal healthcare services in lower SDI regions, aiming to enhance the accessibility and quality of prenatal screening, skilled birth attendance, and comprehensive perinatal care [12, 16–19]. Strengthening healthcare infrastructure and workforce capacity—through expanding NICUs and training providers—is essential to optimize neonatal outcomes [33]. Concurrently, improving vital registration and cause-of-death surveillance systems to enhance data completeness and accuracy is essential to reduce underreporting and misclassification, ensuring robust monitoring of neonatal disease burden and intervention effectiveness [30]. Addressing social determinants of health—such as maternal education, nutritional status, sanitation, and socioeconomic disparities—through multisectoral integrated interventions is crucial to fostering an enabling environment for maternal and child health [33]. Furthermore, in light of the vulnerabilities exposed by the COVID-19 pandemic, reinforcing health system resilience is vital to maintain continuity and stability of maternal and neonatal services during future public health emergencies [86].

This study has several limitations. First, data from some LMICs may be subject to underreporting or low quality, potentially affecting the reliability of the estimates. Second, as diagnostic criteria and surveillance measures for neonatal disorders have evolved over time, the estimates may have been influenced by these changes to some extent. Third, the incidence burden in this study was limited to four primary causes, underestimating the total burden of neonatal morbidity. Although additional neonatal disorders—defined by detailed ICD codes—were included as a fifth category in the mortality analysis, this may still fall short of fully capturing all deaths attributable to neonatal disorders. Additionally, although this study provided a preliminary analysis of the impact of the COVID-19 pandemic, detailed effects of the pandemic remain difficult to fully elucidate. Lastly, this study did not conduct an in-depth investigation into factors such as urban-rural disparities, healthcare system reforms, and maternal nutrition, all of which may influence neonatal health. These factors likely have varying impacts across

different regions, and future research are supposed to refine the analysis by integrating individual data.

Conclusions

In conclusion, although global progress has been made in reducing the burden of neonatal disorders, substantial global disparities persist. Sub-Saharan Africa and South Asia remained high-burden regions requiring particular attention. While the global incidence rate of neonatal disorders declined prior to the COVID-19 pandemic, it began to increase during the pandemic period. In contrast, the mortality rate continued to decline, and in some regions even at a faster pace than before, which may reflect improvements in neonatal care and effective health policy interventions, as well as potential underreporting or misclassification of neonatal deaths during the pandemic. Throughout the first two decades of the 21st century, low-SDI countries have shouldered a disproportionate burden of neonatal disorders. International organizations and governments are supposed to strengthen neonatal health interventions and address global and regional health disparities.

Abbreviations

CCI	Concentration index
CI	Confidence interval
COVID-19	Coronavirus disease 2019
EAPC	Estimated annual percentage change
ENAP	Every newborn action plan
GBD	Global burden of disease
IMR	Infant mortality rate
LMICs	Low- and middle-income countries
NMR	Neonatal mortality rate
SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2
SDGs	Sustainable development goals
SDI	Socio-demographic index
U5MR	Under-five mortality
UI	Uncertain interval
UNICEF	United nations international children's emergency fund
WHO	World health organization

Supplementary Information

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Supplementary material 1.

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Authors' contributions

Authors' Contributions: Conceptualization, C.Q., X.G., and J.L.; methodology and analysis, C.Q.; visualization, C.Q.; writing-original draft preparation, C.Q.; review and editing, Z.F., L.Y., and J.L.; supervision, X.S., Y.D., and J.L. All authors have read and agreed to the published version of the manuscript.

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Data availability

All data in the study are available at <https://www.healthdata.org/>.

Declarations

Ethics approval and consent to participate

Not required as this study used secondary data aggregated at both country and global levels.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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